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MORPHOLOGICAL FILTERS

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Masks

- **Masks** are the primary method for removing or selecting specific parts of an image.
- They are binary arrays that indicate whether a value should be included in an analysis.
- Typically, masks are created by applying one or more logical operations to an image.

Raw image

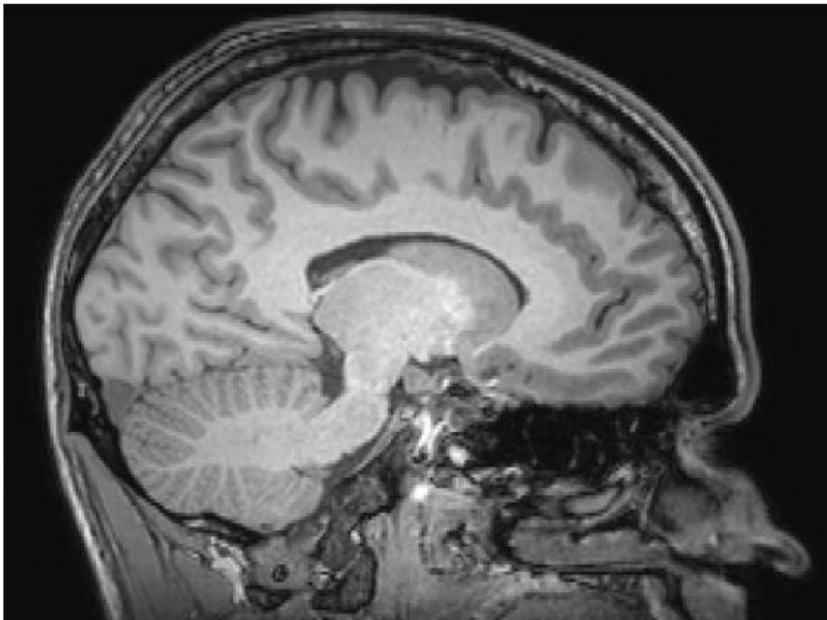


Image mask



Creating masks

- Logical operations result in **True** / **False** at each pixel.

```
1 mat = np.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])  
2 mat
```

```
array([[1, 2, 3],  
       [4, 5, 6],  
       [7, 8, 9]])
```

```
1 mat > 5
```

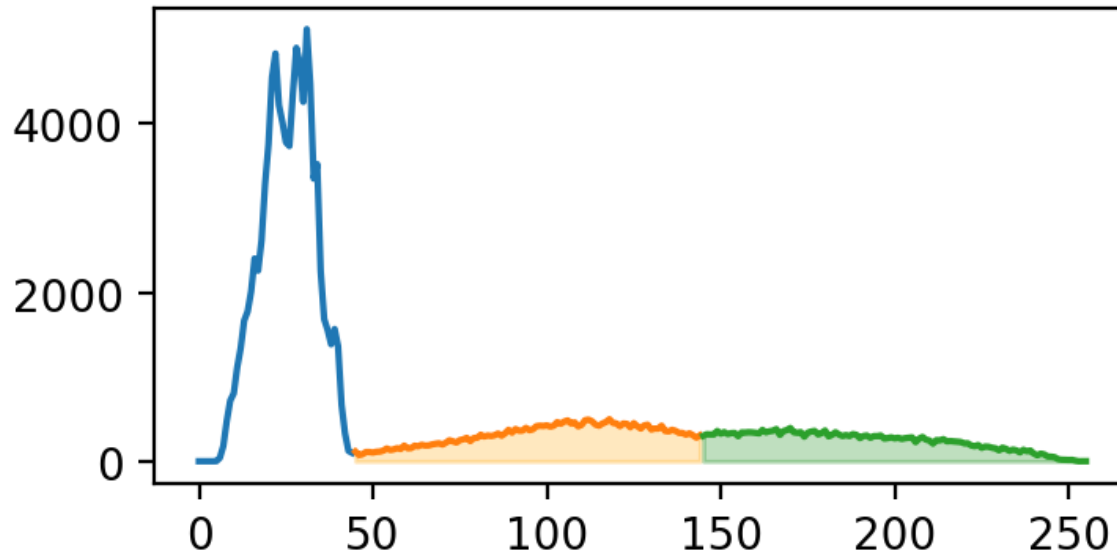
```
array([[False, False, False],  
       [False, False,  True],  
       [ True,  True,  True]])
```

Sample Operations

Operation	Example
Greater	<code>im > 0</code>
Equal to	<code>im == 1</code>
X and Y	<code>(im > 0) & (im < 5)</code>
X or Y	<code>(im > 10) (im < 5)</code>

Thresholding (1) – Create a masks

- Case: We try to use a simple intensity threshold to differentiate between skin and bone in the hand radiograph.
- Below is the histogram of “hand-xray” colored by the segments we will plot.



Thresholding (2) - Create a masks

Instructions (ex):

- Create a bone mask by selecting pixels with intensities greater than or equal to 145.
- Create a skin mask by selecting pixels with intensities greater than or equal to 45 and less than 145.
- Plot the skin and bone masks in grayscale.

```
# Create skin and bone masks
```

```
mask_bone = im >= 145
```

```
mask_skin = (im >= 45) & (im < 145)
```

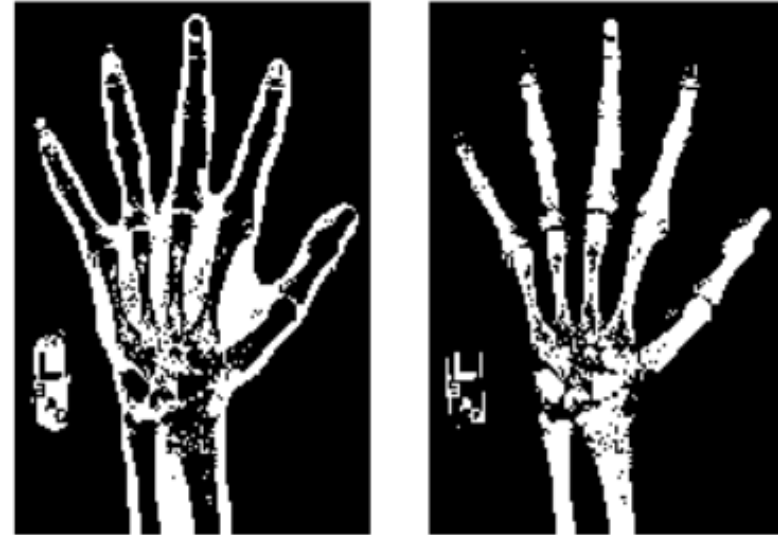
```
# Plot the skin (0) and bone (1) masks
```

```
fig, axes = plt.subplots(1,2)
```

```
axes[0].imshow(mask_skin, cmap='gray')
```

```
axes[1].imshow(mask_bone, cmap='gray')
```

```
plt.show()
```



Thresholding (3) - Apply a mask

- Although masks are binary, they can be applied to images to filter out pixels where the mask is **False**.
- NumPy's **where()** function is a flexible way of applying masks. It takes three arguments:

```
np.where(condition, x, y)
```

- **condition**, **x** and **y** can be either arrays or single values. This allows you to pass through original image values while setting masked values to 0.

Thresholding (4) - Apply a mask

Instructions (ex):

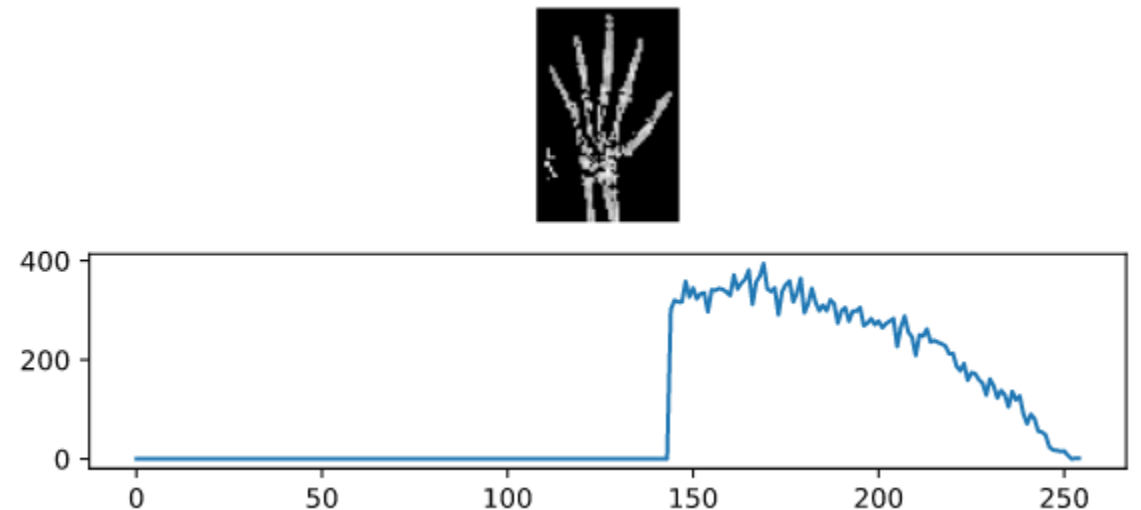
- Create a Boolean bone mask by selecting pixels greater than or equal to 145.
- Apply the mask to your image using `np.where()`. Values not in the mask should be set to 0.
- Create a histogram of the masked image. Use the following arguments to select only non-zero pixels: `min=1`, `max=255`, `bins=255`.

```
# Import SciPy's "ndimage" module
import scipy.ndimage as ndi

# Screen out non-bone pixels from "im"
mask_bone = im >= 145
im_bone = np.where(mask_bone, im, 0)

# Get the histogram of bone intensities
hist = ndi.histogram(im_bone, min=1, max=255, bins=255)

# Plot masked image and histogram
fig, axes = plt.subplots(2,1)
axes[0].imshow(im_bone)
axes[1].plot(hist)
```





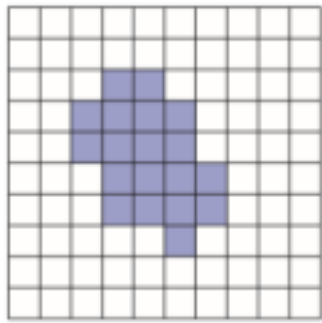
MORPHOLOGICAL FILTERS

Morphological Filters

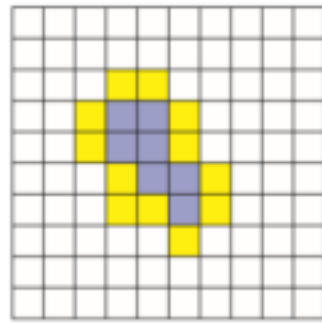
- Morphological filter is used to modify the local structure of the image in the predictable way. It is mostly applied to the binary image (0 and 1) which the value of 0 is called background and 1 is called foreground.
- Morphological filter is applicable not only to binary images but also to grayscale and even color images.
- Operasi morfologi → 2 input : citra biner (dari hasil thresholding) dan kernel.
- Pada morfologi, kernel disebut dengan structuring elements (SE) → matriks berukuran kecil.
- 2 operasi dasar morfologi : Dilasi dan erosi, dapat dikembangkan menjadi opening, closing, dll.

Shrink and grow (1)

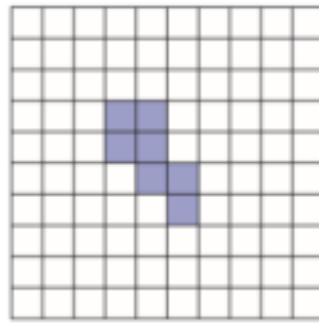
- **Shrinking** remove a layer of pixels from a foreground region around all its borders against the background (a) to (c).



(a)

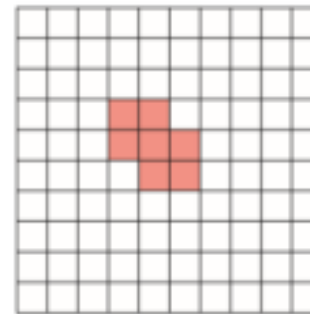


(b)

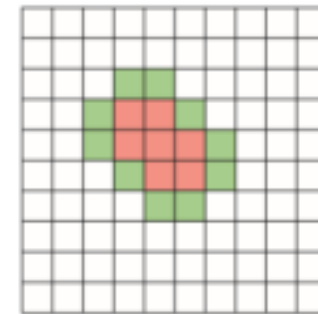


(c)

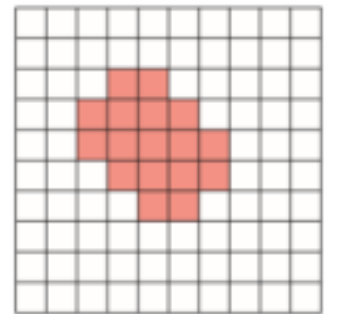
- **Growing** adds a layer of pixels around the border of a foreground region (a) to (c).



(a)



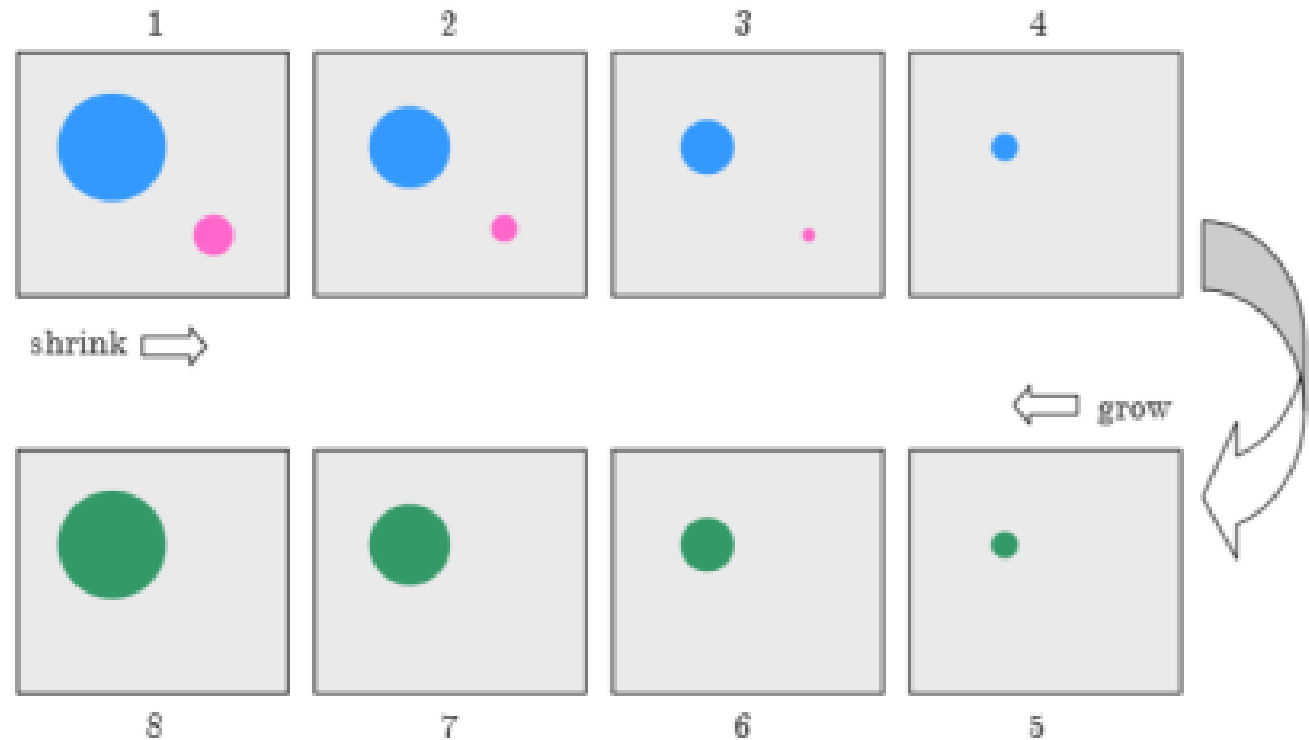
(b)



(c)

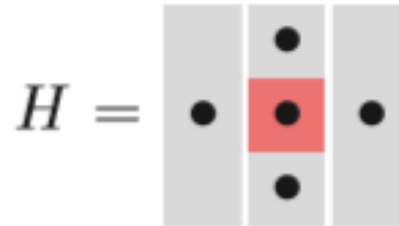
Shrink and grow (2)

- So, shrinking removes small structures, and only larger structures still remain. Remaining structures go back to the original size which larger regions are returned to approximately their original shapes. So, the smaller region will disappear from the image.

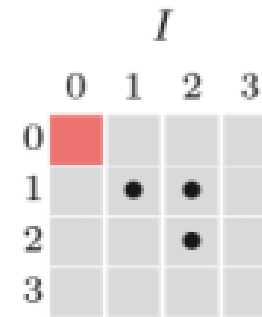


Basic Morphological Operations

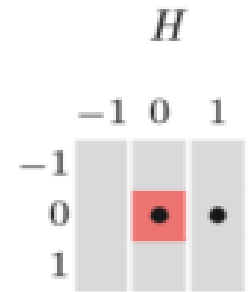
- The structuring element is the properties of a morphological filter. In binary morphology, the structuring element contains only the value 0 and 1. It also has the **hot spot** which mark the original of the filter.
- Binary structuring element (example). 1-elements are marked with •, while 0-elements are empty.



 origin (hot spot)



$$I \equiv \mathcal{Q}_I = \{(1, 1), (2, 1), (2, 2)\}$$

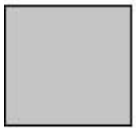


$$H \equiv \mathcal{Q}_H = \{(0, 0), (1, 0)\}$$

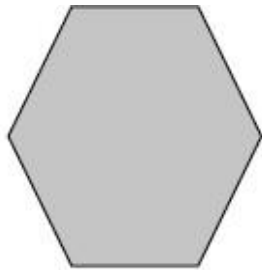
- A binary image I or a structuring element H can each be described as a set of coordinate pairs, $\mathcal{Q}_I$ and $\mathcal{Q}_H$, respectively. The dark shaded element in H marks the coordinate origin (hot spot).

The Structuring Element (1)

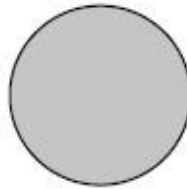
- A structuring element is a shape mask used in the basic morphological operations
- They can be any shape and size that is digitally representable, and each has an origin.



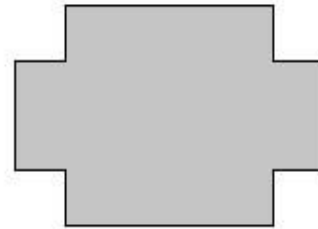
box



hexagon



disk

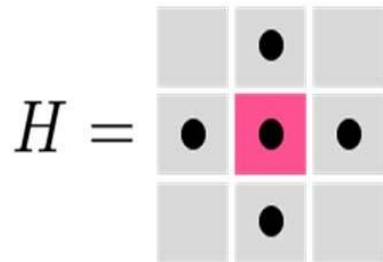


something

The Structuring Element (2)

- Structuring element somewhat similar to a filter
- Contains only 0 and 1 values
- **Hot spot** marks origin of coordinate system of H
- **Example of structuring element:** 1-elements marked with ●, 0-cells are empty.

$$H(i, j) \in \{0, 1\}$$

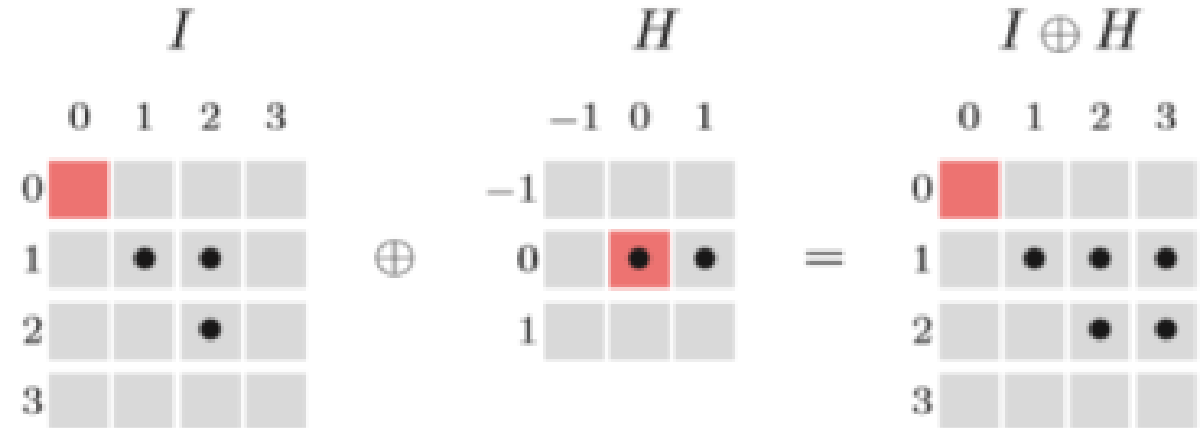


 origin (hot spot)

Dilation (1)

$$A \oplus B = \bigcup_{b \in B} A_b.$$

- It is also called “growing concept” because of the morphological operation.
- Dilation example: The binary image I is subject to dilation with the structuring element H . In the result $I \oplus H$, the structuring element H is replicated at every foreground pixel of the original image I .



$$I \equiv \{(1,1), (2,1), (2,2)\}, \quad H \equiv \{(0,0), (1,0)\}$$

$$I \oplus H \equiv \{ (1,1) + (0,0), (1,1) + (1,0), \\ (2,1) + (0,0), (2,1) + (1,0), \\ (2,2) + (0,0), (2,2) + (1,0) \}$$

Dilation – Example (1)

- For A and B shown below

$$B = \{(0, 0), (1, 1), (-1, 1), (1, -1), (-1, -1)\}$$

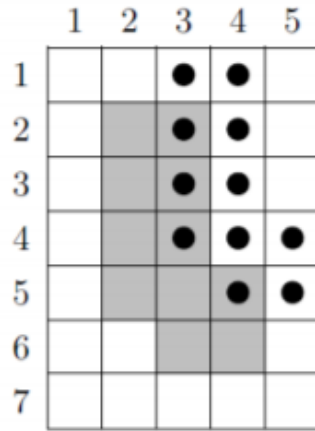
	1	2	3	4	5
1					
2		●	●		
3		●	●		
4		●	●		
5		●	●	●	
6			●	●	
7					

A

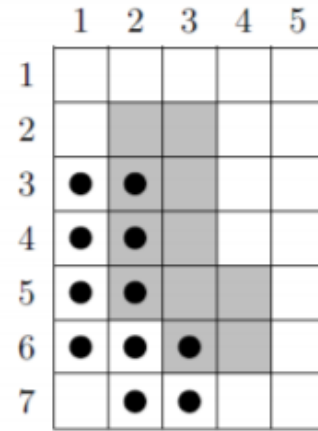
	-1	0	1
-1	●		●
0		●	
1	●		●

B

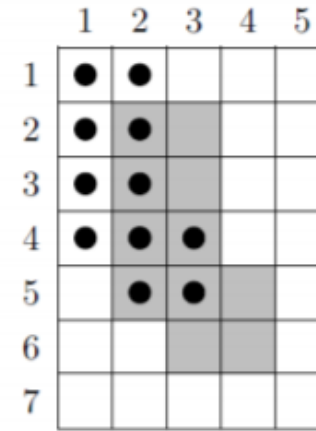
Dilation – Example (2)



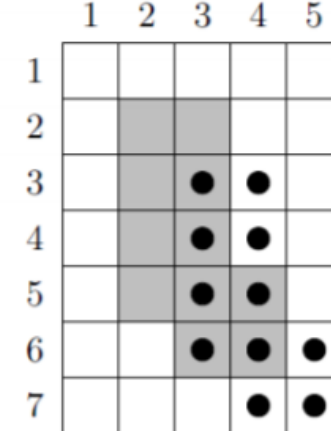
$A_{(-1,1)}$



$A_{(1,-1)}$

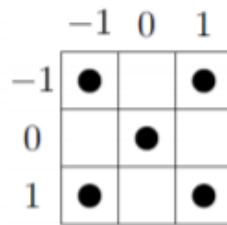


$A_{(-1,-1)}$

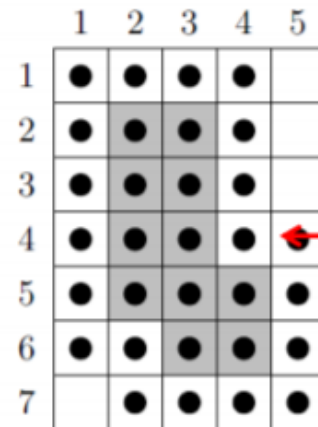


$A_{(1,1)}$

Translation of A
by $(1,1)$



B



$A \oplus B$

Union of all translations

Example: Dilation of a Binary Image

Cross-Correlation Used
To Locate A Known
Target in an Image

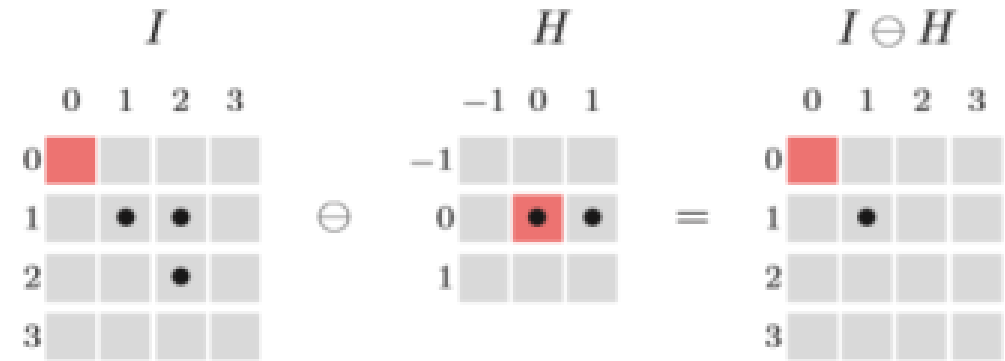
Text Running
In Another
Direction

**Cross-Correlation Used
To Locate A Known
Target in an Image**

**Text Running
In Another
Direction**

Erosion

- It is also called “shrinking concept” because of the morphological operation.
- Erosion example: : The binary image I is subject to erosion with the structuring element H . H is only covered by I when placed at position $p=(1,1)$. Thus the resulting point set contains only the single coordinate $(1,1)$.



$$I \equiv \{(1,1), (2,1), (2,2)\}, H \equiv \{(0,0), (1,0)\}$$

$$I \ominus H \equiv \{(1,1)\} \text{ because}$$

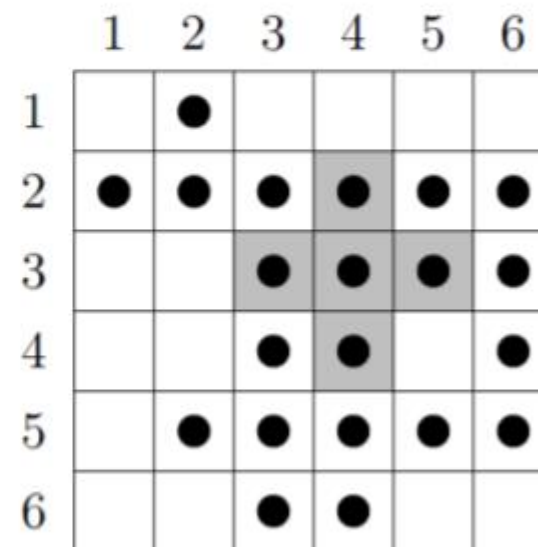
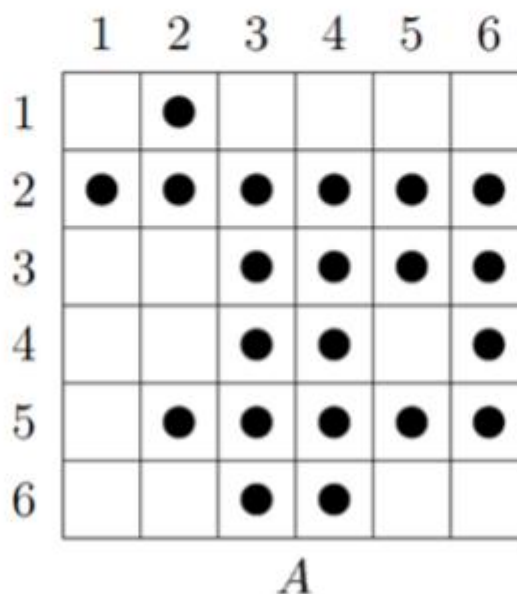
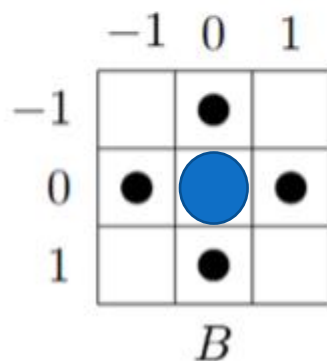
$$(1,1) + (0,0) = (1,1) \in I \quad \text{and} \quad (1,1) + (1,0) = (2,1) \in I$$

Erosion – Example (1)

- Given sets A and B , the **erosion of A by B**

$$A \ominus B = \{w : B_w \subseteq A\}.$$

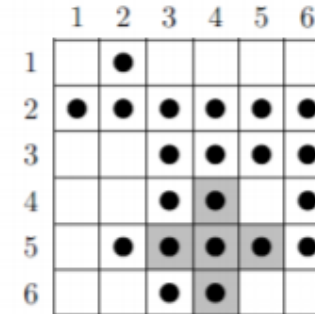
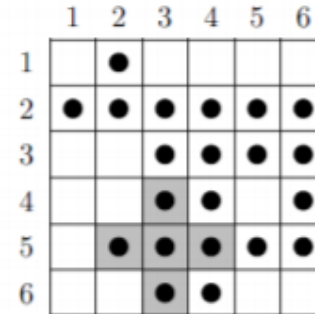
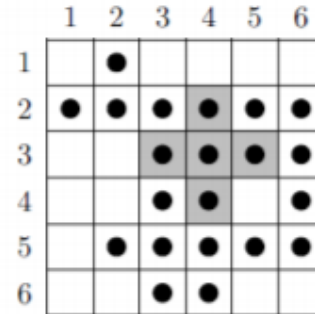
- Find all occurrences of B in A



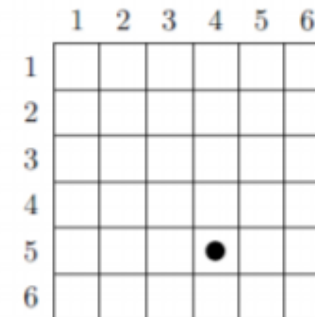
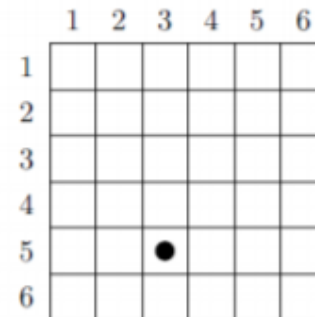
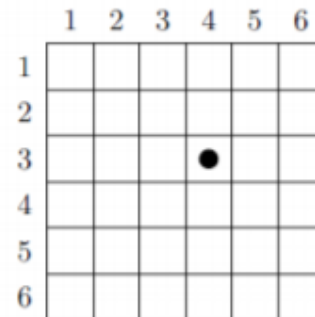
Example: 1 occurrence
of B in A

Erosion – Example (2)

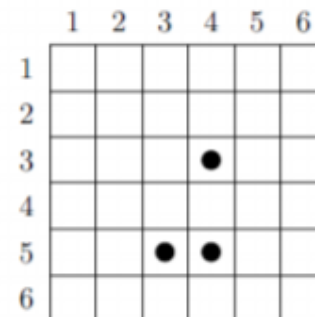
All occurrences
of B in A



For each
occurrences
Mark center of B

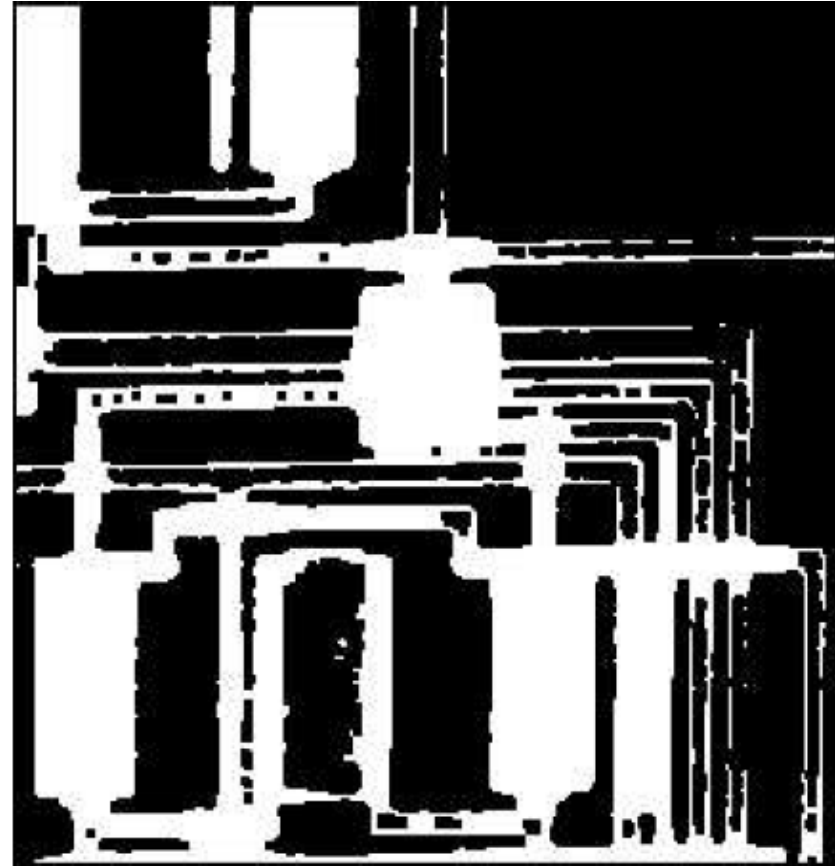
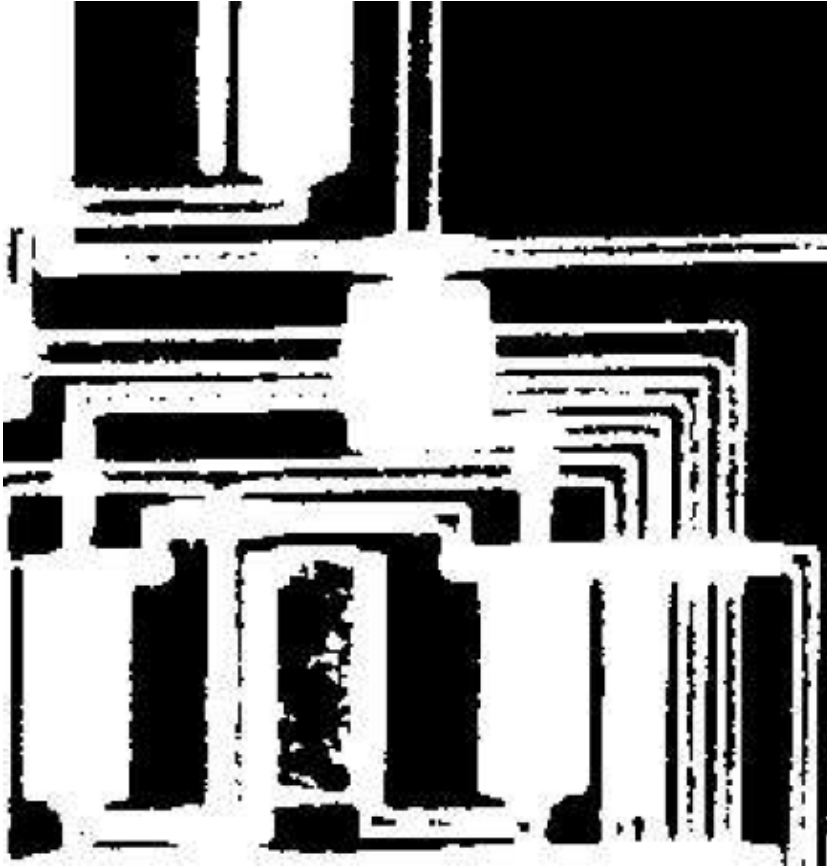


Erosion: union
of center of all
occurrences of
 B in A



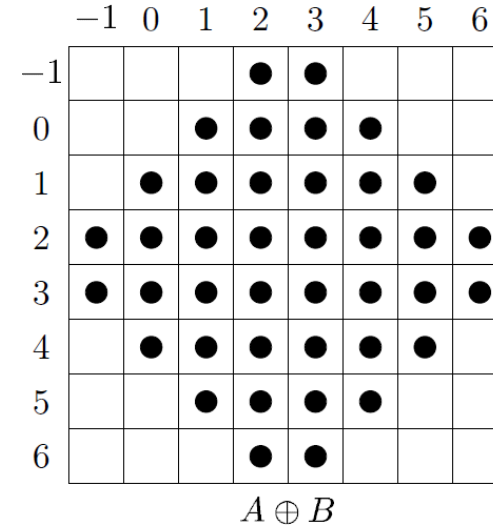
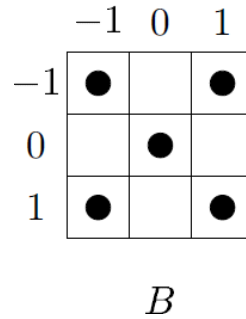
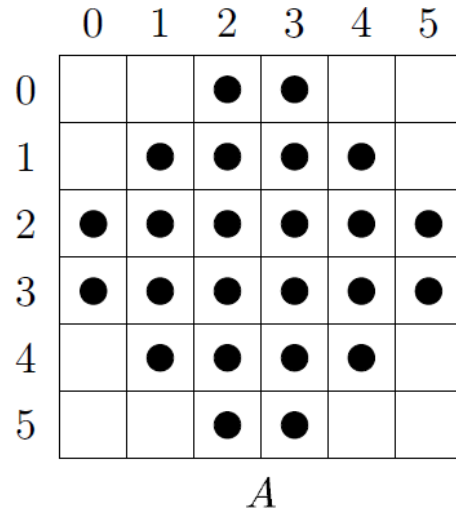
$$A \ominus B$$

Example: Erosion of Binary Image

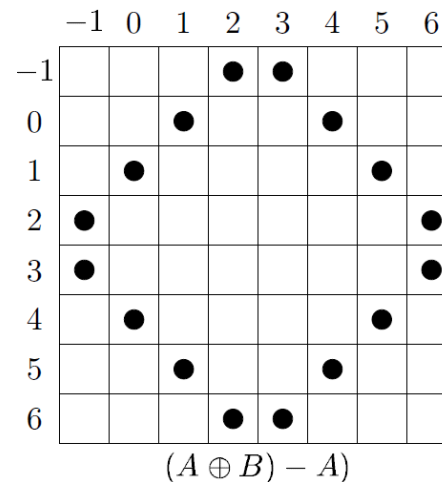


An Application: Boundary Detection

- Given an image A and structuring element B

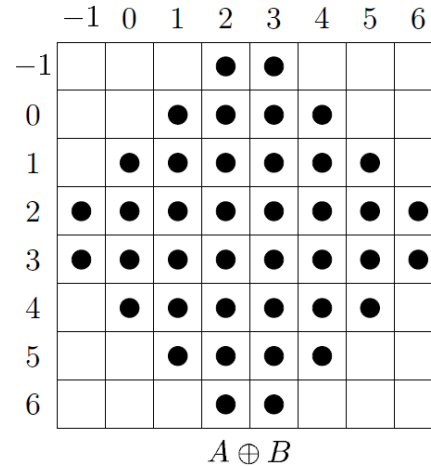
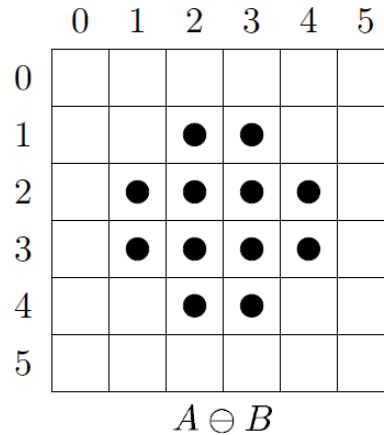


We can define **external boundary**



An Application: Boundary Detection

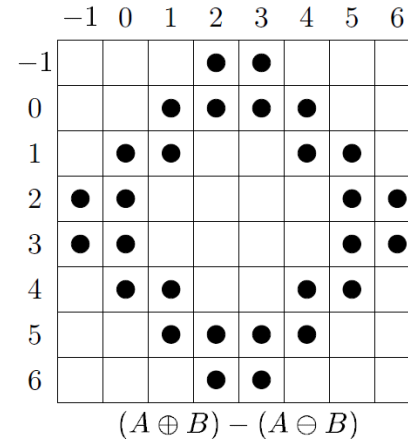
- **Dilation of image A - erosion image A** (by structuring element B)



- We can define **morphological gradient**

$$(A \oplus B) - (A \ominus B)$$

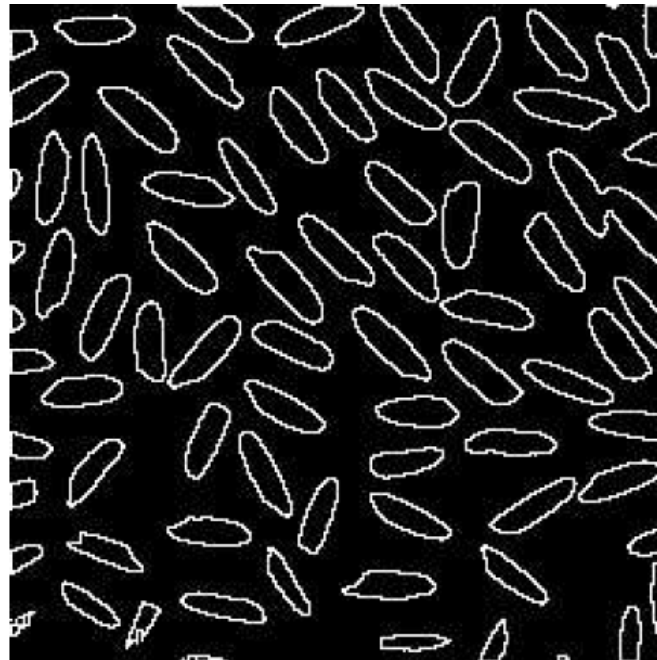
Morphological gradient = Dilation - erosion



Example: Internal Boundary of Binary Image

- We can also define **internal boundary** as

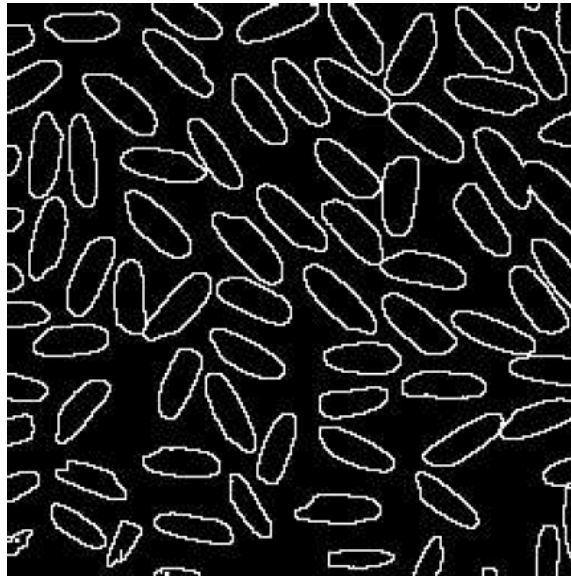
$$A - (A \ominus B)$$



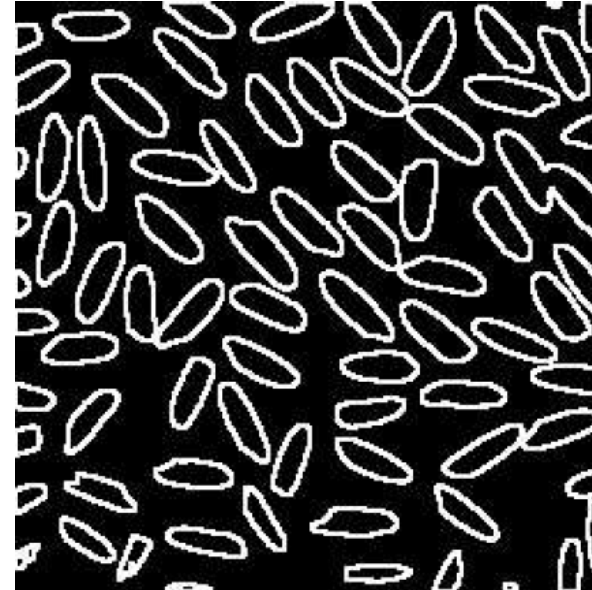
Example: External Boundary and Morphological Gradient



Image



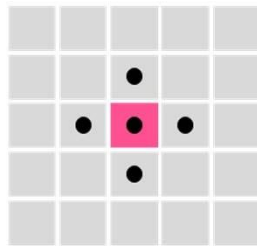
External Boundary



Morphological
Gradient

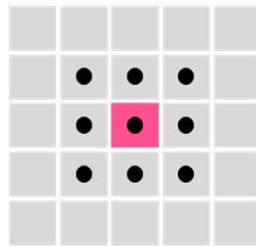
Designing Morphological Filters

- A morphological filter is specified by:
 - Type of operation (e.g. dilation, erosion)
 - Contents of structuring element



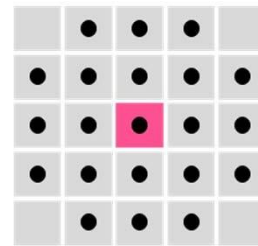
(a)

4-neighborhood



(b)

8-neighborhood



(c)

Small Disk
(circular)

- In practice, quasi-circular shaped structuring elements used
- Dilation with circular structuring of radius r adds thickness r
- Erosion with circular structuring of radius r removes thickness r

Dilation

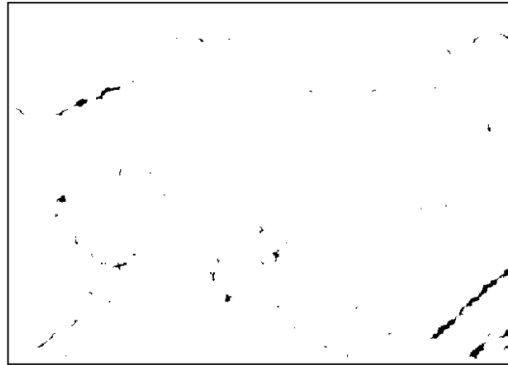
Erosion



$r = 1.0$



$r = 2.5$



$r = 5.0$

Example: Dilation and Erosion

Composite operation

- Dilation and erosion are often used together in composite operations. They are called **opening** and **closing**.

Opening

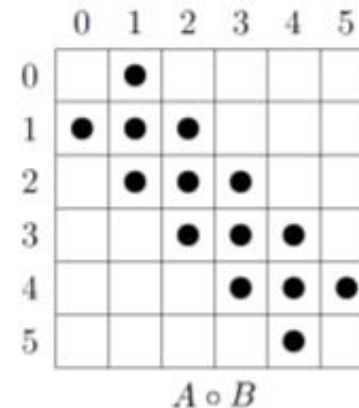
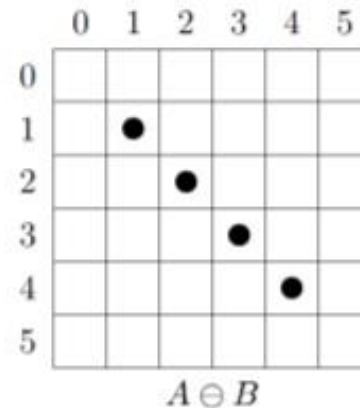
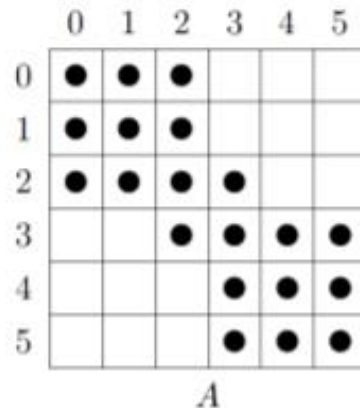
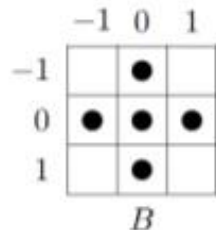
- Opening and closing: operations built on dilation and erosion
- Opening of A by structuring element B

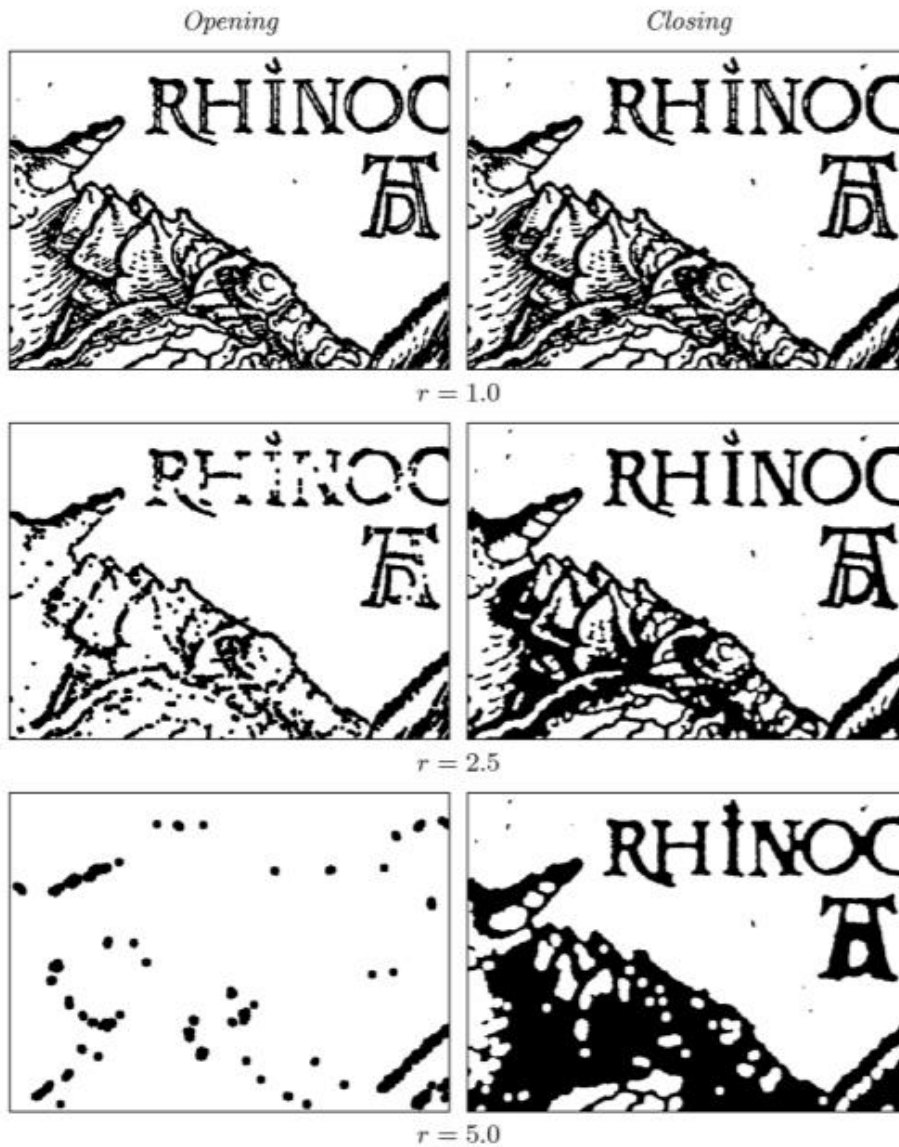
$$A \circ B = (A \ominus B) \oplus B.$$

- i.e. opening = erosion followed by dilation. Alternatively

$$A \circ B = \cup \{B_w : B_w \subseteq A\}.$$

- i.e. Opening = union of all translations of B that fit in A
- **Note:** Opening includes all of B , erosion includes just $(0,0)$ of B

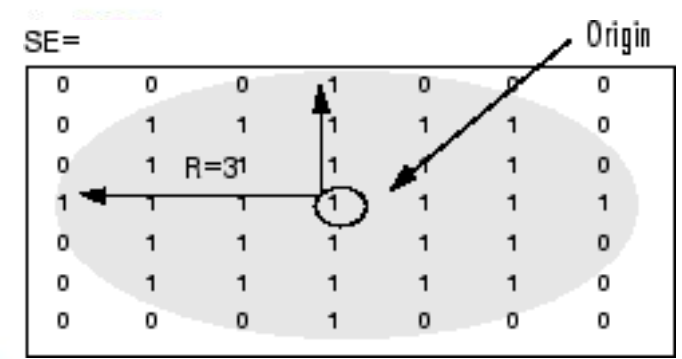




Binary opening and closing with disk-shaped Structuring elements of radius $r = 1.0, 2.5, 5.0$

Opening

- All foreground structures smaller than structuring element are eliminated by first step (erosion)
- Remaining structures smoothed by next step (dilation) then grown back to their original size



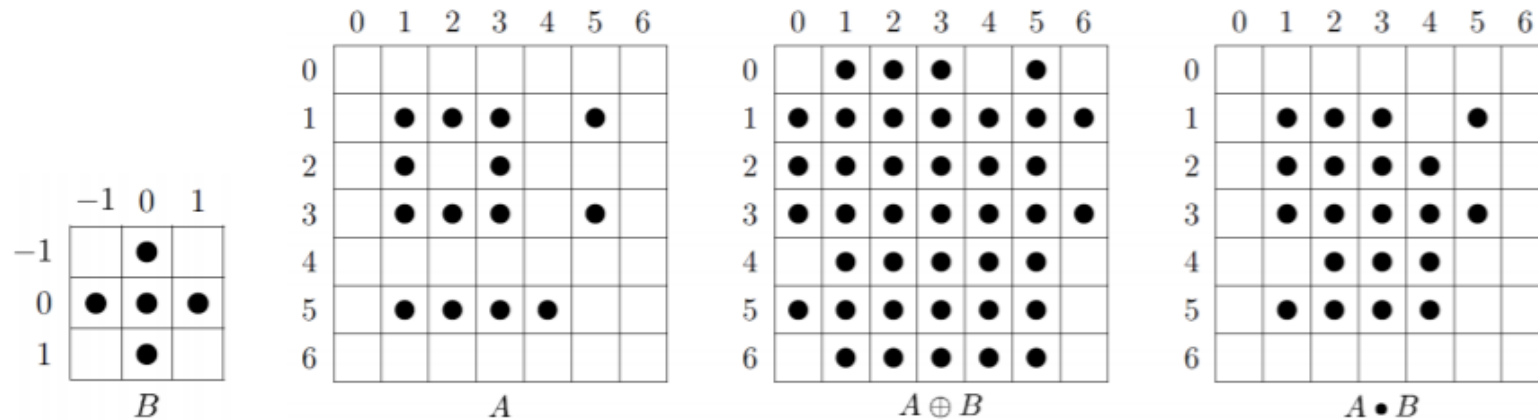
Opening tends to smooth an image, break narrow joins, and remove thin protrusions.

Closing

- Closing of A by structuring element B

$$A \bullet B = (A \oplus B) \ominus B.$$

- i.e. closing = dilation followed by erosion



Closing tends to:

- Smooth an image
- Fuse narrow breaks and thin gulfs
- Eliminates small holes.

Computation (1)

Imperfect masks can be tuned through the addition and subtraction of pixels. SciPy includes several useful methods for accomplishing these ends. These include:

- `binary_dilation` : Add pixels along edges
- `binary_erosion` : Remove pixels along edges
- `binary_opening` : Erode then dilate, "opening" areas near edges
- `binary_closing` : Dilate then erode, "filling in" holes

For this exercise, create a bone mask then tune it to include additional pixels.

For the remaining exercises, we have run the following import for you:

```
import scipy.ndimage as ndi
```

Computation (2) - Tune a mask

Instructions (ex):

- Create a bone by selecting pixels from `im` that are greater than or equal to `145`.
- Use `ndi.binary_dilation()` to increase the size of `mask_bone`. Set the number of `iterations` to `5` to perform the dilation multiple times.
- Use `ndi.binary_closing()` to fill in holes in `mask_bone`. Set the number of `iterations` to `5` to holes up to 10 pixels wide.
- Plot the original and tuned masks.

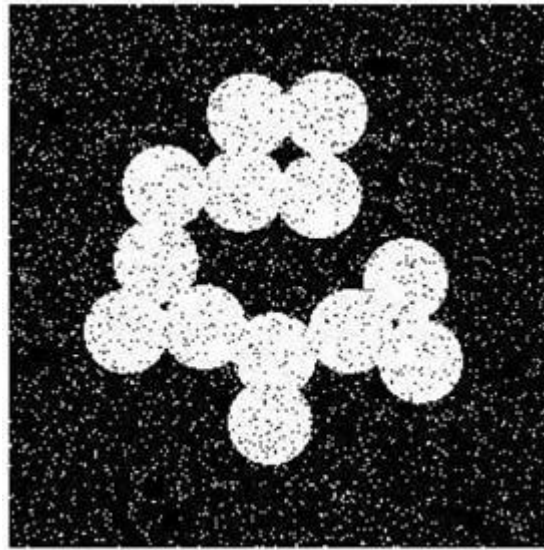
```
# Create and tune bone mask
mask_bone = np.where(im >=145, 1, 0)
mask_dilate = ndi.binary_dilation(mask_bone,
iterations=5)
mask_closed = ndi.binary_closing(mask_bone,
iterations=5)

# Plot masked images
fig, axes = plt.subplots(1,3)
axes[0].imshow(mask_bone)
axes[1].imshow(mask_dilate)
axes[2].imshow(mask_closed)
```



Noise Removal: Morphological Filtering

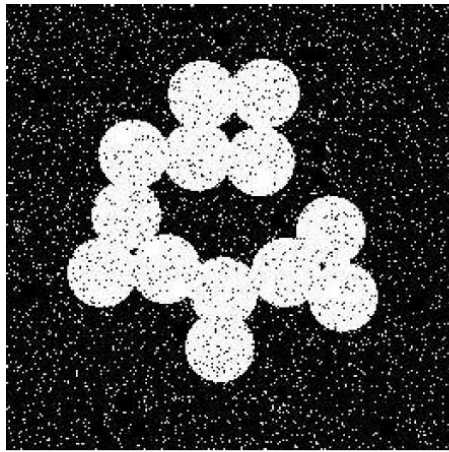
- Suppose A is image corrupted by impulse noise (some black, some white pixels, shown in (a) below)



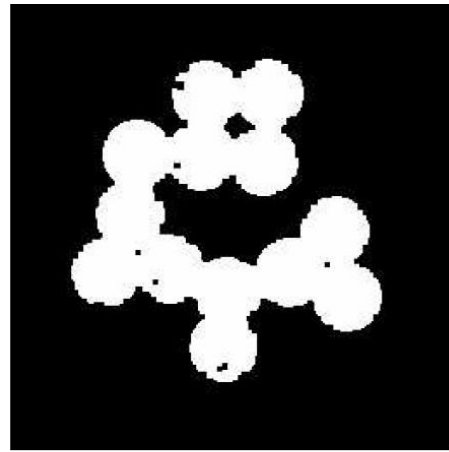
(a)

- $A \ominus B$ removes single black pixels, but enlarges holes
- We can fill holes by dilating twice $((A \ominus B) \oplus B) \oplus B$.

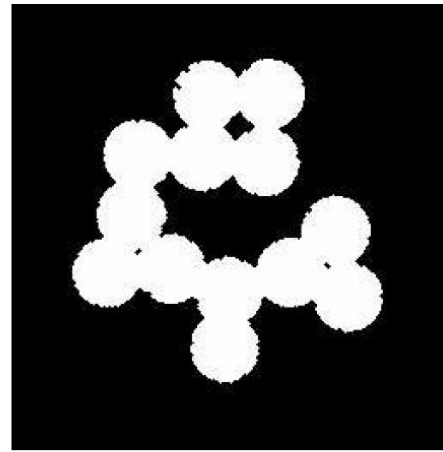
Noise Removal: Morphological Filtering



(a)



(b)



(c)

(b) Filter once
(c) Filter Twice

- First dilation returns the holes to their original size
- Second dilation removes the holes but enlarges objects in image
- To reduce them to their correct size, perform a final erosion:

$$(((A \ominus B) \oplus B) \oplus B) \ominus B.$$

- Inner 2 operations = opening, Outer 2 operations = closing.
- This noise removal method = opening followed by closing
 $(A \circ B) \bullet B).$

References

- Wilhelm Burger and Mark J. Burge, Digital Image Processing, Springer, 2008
- University of Utah, CS 4640: Image Processing Basics, Spring 2012
- Rutgers University, CS 334, Introduction to Imaging and Multimedia, Fall 2012
- Gonzales and Woods, Digital Image Processing (3rd edition), Prentice Hall